

Personal Astro Almanac 2027

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Sun/Twilight (sample)

date	dusk_astronomical_local	dawn_astronomical_local
2027-01-01	2027-01-01T19:14:34.287309+10:00	2027-01-01T04:27:47.379546+10:00
2027-01-02	2027-01-02T19:14:47.646766+10:00	2027-01-02T04:28:29.671031+10:00
2027-01-03	2027-01-03T19:14:59.391090+10:00	2027-01-03T04:29:12.931129+10:00
2027-01-04	2027-01-04T19:15:09.506803+10:00	2027-01-04T04:29:57.114585+10:00
2027-01-05	2027-01-05T19:15:17.982204+10:00	2027-01-05T04:30:42.175583+10:00
2027-01-06	2027-01-06T19:15:24.807352+10:00	2027-01-06T04:31:28.067842+10:00
2027-01-07	2027-01-07T19:15:29.974059+10:00	2027-01-07T04:32:14.744717+10:00
2027-01-08	2027-01-08T19:15:33.475864+10:00	2027-01-08T04:33:02.159303+10:00
2027-01-09	2027-01-09T19:15:35.308017+10:00	2027-01-09T04:33:50.264529+10:00
2027-01-10	2027-01-10T19:15:35.467447+10:00	2027-01-10T04:34:39.013266+10:00
2027-01-11	2027-01-11T19:15:33.952740+10:00	2027-01-11T04:35:28.358421+10:00
2027-01-12	2027-01-12T19:15:30.764098+10:00	2027-01-12T04:36:18.253042+10:00
2027-01-13	2027-01-13T19:15:25.903312+10:00	2027-01-13T04:37:08.650409+10:00
2027-01-14	2027-01-14T19:15:19.373722+10:00	2027-01-14T04:37:59.504135+10:00
2027-01-15	2027-01-15T19:15:11.180176+10:00	2027-01-15T04:38:50.768256+10:00
2027-01-16	2027-01-16T19:15:01.328993+10:00	2027-01-16T04:39:42.397321+10:00
2027-01-17	2027-01-17T19:14:49.827917+10:00	2027-01-17T04:40:34.346484+10:00
2027-01-18	2027-01-18T19:14:36.686076+10:00	2027-01-18T04:41:26.571582+10:00
2027-01-19	2027-01-19T19:14:21.913935+10:00	2027-01-19T04:42:19.029218+10:00
2027-01-20	2027-01-20T19:14:05.523250+10:00	2027-01-20T04:43:11.676842+10:00

Moon Phase/Illumination (sample)

date	moon_phase_index	moon_illumination_fraction
2027-01-01	22.589	0.4531471156174623
2027-01-02	23.444555555555553	0.363803645471525
2027-01-03	24.377888888888889	0.27156956997927967
2027-01-04	25.233444444444444	0.194826534942508
2027-01-05	26.089	0.12816811048193671
2027-01-06	26.866777777777777	0.07818171055259243
2027-01-07	27.722333333333333	0.03655200636702338
2027-01-08	0.5778888888888889	0.0037748265870695685
2027-01-09	1.4334444444444445	0.023075341738515887
2027-01-10	2.289	0.05813607199204657

date	moon_phase_index	moon_illumination_fraction
2027-01-11	3.1445555555555558	0.10779841788994848
2027-01-12	4.000111111111111	0.17042126224470144
2027-01-13	4.8556666666666666	0.24393520168149796
2027-01-14	5.711222222222223	0.3259109311131388
2027-01-15	6.6445555555555556	0.42180095644903964
2027-01-16	7.6556666666666668	0.5290286711629395
2027-01-17	8.589	0.6269325281569588
2027-01-18	9.600111111111111	0.7272482806899407
2027-01-19	10.689	0.8234409312316604
2027-01-20	11.700111111111111	0.8973827920826103

Moonrise/Moonset (sample)

date	moonrise_local	moonset_local
2027-01-01		2027-01-01T13:06:00+10:00
2027-01-02	2027-01-02T00:21:00+10:00	2027-01-02T14:04:00+10:00
2027-01-03	2027-01-03T00:57:00+10:00	2027-01-03T15:01:00+10:00
2027-01-04	2027-01-04T01:36:00+10:00	2027-01-04T15:58:00+10:00
2027-01-05	2027-01-05T02:20:00+10:00	2027-01-05T16:53:00+10:00
2027-01-06	2027-01-06T03:09:00+10:00	2027-01-06T17:44:00+10:00
2027-01-07	2027-01-07T04:02:00+10:00	2027-01-07T18:31:00+10:00
2027-01-08	2027-01-08T04:57:00+10:00	2027-01-08T19:13:00+10:00
2027-01-09	2027-01-09T05:52:00+10:00	2027-01-09T19:50:00+10:00
2027-01-10	2027-01-10T06:48:00+10:00	2027-01-10T20:24:00+10:00
2027-01-11	2027-01-11T07:42:00+10:00	2027-01-11T20:55:00+10:00
2027-01-12	2027-01-12T08:36:00+10:00	2027-01-12T21:24:00+10:00
2027-01-13		2027-01-13T21:52:00+10:00
2027-01-14	2027-01-14T10:24:00+10:00	2027-01-14T22:21:00+10:00
2027-01-15	2027-01-15T11:21:00+10:00	2027-01-15T22:53:00+10:00
2027-01-16	2027-01-16T12:20:00+10:00	2027-01-16T23:29:00+10:00
2027-01-17	2027-01-17T13:24:00+10:00	
2027-01-18	2027-01-18T14:31:00+10:00	2027-01-18T00:10:00+10:00
2027-01-19	2027-01-19T15:40:00+10:00	2027-01-19T01:00:00+10:00
2027-01-20	2027-01-20T16:47:00+10:00	2027-01-20T02:00:00+10:00

Moon-Free Dark Windows (sample)

date	start_local	end_local	duration_minutes	moon_illumination_fraction
2027-01-01	2027-01-01T19:14:34.287309+10:00	2027-01-02T04:27:47.379546+10:00	553.2	0.4
2027-01-02	2027-01-02T19:14:47.646766+10:00	2027-01-03T04:28:29.671031+10:00	553.7	0.3
2027-01-03	2027-01-03T19:14:59.391090+10:00	2027-01-04T04:29:12.931129+10:00	554.2	0.2
2027-01-04	2027-01-04T19:15:09.506803+10:00	2027-01-05T04:29:57.114585+10:00	554.8	0.1
2027-01-05	2027-01-05T19:15:17.982204+10:00	2027-01-06T04:30:42.175583+10:00	555.4	0.1

date	start_local	end_local	duration_minutes	max
2027-01-06	2027-01-06T19:15:24.807352+10:00	2027-01-07T04:31:28.067842+10:00	556.1	0.0
2027-01-07	2027-01-07T19:15:29.974059+10:00	2027-01-08T04:32:14.744717+10:00	556.7	0.0
2027-01-08	2027-01-08T19:15:33.475864+10:00	2027-01-09T04:33:02.159303+10:00	557.5	0.0
2027-01-09	2027-01-09T19:15:35.308017+10:00	2027-01-10T04:33:50.264529+10:00	558.2	0.0
2027-01-10	2027-01-10T19:15:35.467447+10:00	2027-01-11T04:34:39.013266+10:00	559.1	0.0
2027-01-11	2027-01-11T19:15:33.952740+10:00	2027-01-12T04:35:28.358421+10:00	559.9	0.1
2027-01-12	2027-01-12T19:15:30.764098+10:00	2027-01-13T04:36:18.253042+10:00	560.8	0.1
2027-01-13	2027-01-13T19:15:25.903312+10:00	2027-01-14T04:37:08.650409+10:00	561.7	0.2
2027-01-14	2027-01-14T22:21:00+10:00	2027-01-15T04:37:59.504135+10:00	377.0	0.3
2027-01-15	2027-01-15T22:53:00+10:00	2027-01-16T04:38:50.768256+10:00	345.8	0.4
2027-01-16	2027-01-16T23:29:00+10:00	2027-01-17T04:39:42.397321+10:00	310.7	0.5
2027-01-17	2027-01-17T19:14:49.827917+10:00	2027-01-18T04:40:34.346484+10:00	565.7	0.6
2027-01-18	2027-01-19T00:10:00+10:00	2027-01-19T04:41:26.571582+10:00	271.4	0.7
2027-01-19	2027-01-20T01:00:00+10:00	2027-01-20T04:42:19.029218+10:00	222.3	0.8
2027-01-20	2027-01-21T02:00:00+10:00	2027-01-21T04:43:11.676842+10:00	163.2	0.8

Milky Way Core Visibility (sample)

date	start_local	end_local	duration_minutes	max
2027-01-16	2027-01-16T04:33:00+10:00	2027-01-16T04:33:00+10:00	10.0	20.0
2027-01-17	2027-01-17T04:29:00+10:00	2027-01-17T04:39:00+10:00	20.0	22.0
2027-01-18	2027-01-18T04:24:49.827917+10:00	2027-01-18T04:34:49.827917+10:00	20.0	22.0
2027-01-19	2027-01-19T04:20:00+10:00	2027-01-19T04:40:00+10:00	30.0	24.0
2027-01-20	2027-01-20T04:20:00+10:00	2027-01-20T04:40:00+10:00	30.0	25.0
2027-01-21	2027-01-21T04:20:00+10:00	2027-01-21T04:40:00+10:00	30.0	26.0
2027-01-22	2027-01-22T04:07:00+10:00	2027-01-22T04:37:00+10:00	40.0	26.0
2027-01-28	2027-01-28T03:51:26.706593+10:00	2027-01-28T04:41:26.706593+10:00	60.0	32.0
2027-02-01	2027-02-01T03:29:22.917266+10:00	2027-02-01T04:49:22.917266+10:00	90.0	37.0
2027-02-02	2027-02-02T03:28:48.427573+10:00	2027-02-02T04:48:48.427573+10:00	90.0	37.0
2027-02-03	2027-02-03T03:28:12.568510+10:00	2027-02-03T04:48:12.568510+10:00	90.0	38.0
2027-02-04	2027-02-04T03:17:35.365053+10:00	2027-02-04T04:47:35.365053+10:00	100.0	39.0
2027-02-05	2027-02-05T03:16:56.842792+10:00	2027-02-05T04:46:56.842792+10:00	100.0	39.0
2027-02-06	2027-02-06T03:16:17.027888+10:00	2027-02-06T04:56:17.027888+10:00	110.0	42.0
2027-02-07	2027-02-07T03:05:35.947036+10:00	2027-02-07T04:55:35.947036+10:00	120.0	43.0
2027-02-08	2027-02-08T03:04:53.627426+10:00	2027-02-08T04:54:53.627426+10:00	120.0	44.0
2027-02-09	2027-02-09T03:04:10.096706+10:00	2027-02-09T04:54:10.096706+10:00	120.0	44.0
2027-02-10	2027-02-10T02:53:25.382947+10:00	2027-02-10T04:53:25.382947+10:00	130.0	45.0
2027-02-11	2027-02-11T02:52:39.514604+10:00	2027-02-11T04:52:39.514604+10:00	130.0	46.0
2027-02-12	2027-02-12T02:51:52.520487+10:00	2027-02-12T05:01:52.520487+10:00	140.0	49.0

Milky Way Monthly Summary (sample)

month	window_count	excellent_count	best_altitude_deg
2027-01	8	4	32.08
2027-02	21	21	62.63
2027-03	22	22	88.5
2027-04	24	23	88.55
2027-05	26	26	88.55
2027-06	27	27	88.55
2027-07	25	25	88.55
2027-08	24	23	88.55
2027-09	20	20	88.55
2027-10	20	20	73.04
2027-11	13	8	42.33

Planet Visibility Daily (sample)

date	planet	best_time_local	max_altitude_deg	visibility_rating
2027-01-01	Mercury	2027-01-01T12:00:00+10:00	86.5	excellent
2027-01-01	Venus	2027-01-01T08:30:00+10:00	77.81	excellent
2027-01-01	Mars	2027-01-01T04:00:00+10:00	51.59	excellent
2027-01-01	Jupiter	2027-01-01T03:00:00+10:00	48.92	excellent
2027-01-01	Saturn	2027-01-01T17:30:00+10:00	61.35	excellent
2027-01-01	Uranus	2027-01-01T21:00:00+10:00	42.0	good
2027-01-01	Neptune	2027-01-01T17:00:00+10:00	62.88	excellent
2027-01-02	Mercury	2027-01-02T12:00:00+10:00	86.83	excellent
2027-01-02	Venus	2027-01-02T08:30:00+10:00	78.05	excellent
2027-01-02	Mars	2027-01-02T04:00:00+10:00	51.58	excellent
2027-01-02	Jupiter	2027-01-02T03:00:00+10:00	48.9	excellent
2027-01-02	Saturn	2027-01-02T17:30:00+10:00	61.39	excellent
2027-01-02	Uranus	2027-01-02T21:00:00+10:00	42.03	good
2027-01-02	Neptune	2027-01-02T17:00:00+10:00	62.97	excellent
2027-01-03	Mercury	2027-01-03T12:00:00+10:00	87.01	excellent
2027-01-03	Venus	2027-01-03T08:30:00+10:00	78.28	excellent
2027-01-03	Mars	2027-01-03T04:00:00+10:00	51.55	excellent
2027-01-03	Jupiter	2027-01-03T03:00:00+10:00	48.86	excellent
2027-01-03	Saturn	2027-01-03T17:30:00+10:00	61.4	excellent
2027-01-03	Uranus	2027-01-03T21:00:00+10:00	42.03	good

Planet Visibility Monthly Summary (sample)

month	planet	best_date	best_altitude_deg	rating
2027-01	Jupiter	2027-01-01	48.92	excellent
2027-01	Mars	2027-01-01	51.59	excellent

month	planet	best_date	best_altitude_deg	rating
2027-01	Mercury	2027-01-03	87.01	excellent
2027-01	Neptune	2027-01-04	63.06	excellent
2027-01	Saturn	2027-01-03	61.4	excellent
2027-01	Uranus	2027-01-25	42.12	good
2027-01	Venus	2027-01-31	82.97	excellent
2027-02	Jupiter	2027-02-01	47.73	excellent
2027-02	Mars	2027-02-01	49.69	excellent
2027-02	Mercury	2027-02-28	74.17	excellent
2027-02	Neptune	2027-02-04	62.78	excellent
2027-02	Saturn	2027-02-04	60.45	excellent
2027-02	Uranus	2027-02-09	42.13	good
2027-02	Venus	2027-02-10	83.76	excellent
2027-03	Jupiter	2027-03-02	46.5	excellent
2027-03	Mars	2027-03-01	45.77	excellent
2027-03	Mercury	2027-03-08	75.82	excellent
2027-03	Neptune	2027-03-01	62.46	excellent
2027-03	Saturn	2027-03-01	59.4	excellent
2027-03	Uranus	2027-03-04	42.08	good

Jupiter Moons Relative Offsets (sample)

system	date	datetime_utc	datetime_local	moon	dra_arcsec	ddec_arcsec
Jupiter	2027-12-31	2027-12-31T00:00:00	2027-12-31T10:00:00+10:00	Io	16.869	-3.744
Jupiter	2027-12-31	2027-12-31T01:00:00	2027-12-31T11:00:00+10:00	Io	1.762	3.492
Jupiter	2027-12-31	2027-12-31T02:00:00	2027-12-31T12:00:00+10:00	Io	-13.344	10.692
Jupiter	2027-12-31	2027-12-31T03:00:00	2027-12-31T13:00:00+10:00	Io	-28.235	17.676
Jupiter	2027-12-31	2027-12-31T04:00:00	2027-12-31T14:00:00+10:00	Io	-42.443	24.228
Jupiter	2027-12-31	2027-12-31T05:00:00	2027-12-31T15:00:00+10:00	Io	-55.715	30.24
Jupiter	2027-12-31	2027-12-31T06:00:00	2027-12-31T16:00:00+10:00	Io	-67.765	35.604
Jupiter	2027-12-31	2027-12-31T07:00:00	2027-12-31T17:00:00+10:00	Io	-78.303	40.176
Jupiter	2027-12-31	2027-12-31T08:00:00	2027-12-31T18:00:00+10:00	Io	-87.152	43.848
Jupiter	2027-12-31	2027-12-31T09:00:00	2027-12-31T19:00:00+10:00	Io	-94.058	46.548
Jupiter	2027-12-31	2027-12-31T10:00:00	2027-12-31T20:00:00+10:00	Io	-98.878	48.24
Jupiter	2027-12-31	2027-12-31T11:00:00	2027-12-31T21:00:00+10:00	Io	-101.503	48.888
Jupiter	2027-12-31	2027-12-31T12:00:00	2027-12-31T22:00:00+10:00	Io	-101.935	48.384
Jupiter	2027-12-31	2027-12-31T13:00:00	2027-12-31T23:00:00+10:00	Io	-100.029	46.872
Jupiter	2027-12-31	2027-12-31T14:00:00	2028-01-01T00:00:00+10:00	Io	-96.0	44.316
Jupiter	2027-12-31	2027-12-31T15:00:00	2028-01-01T01:00:00+10:00	Io	-89.85	40.788
Jupiter	2027-12-31	2027-12-31T16:00:00	2028-01-01T02:00:00+10:00	Io	-81.685	36.36
Jupiter	2027-12-31	2027-12-31T17:00:00	2028-01-01T03:00:00+10:00	Io	-71.793	31.104
Jupiter	2027-12-31	2027-12-31T18:00:00	2028-01-01T04:00:00+10:00	Io	-60.283	25.2
Jupiter	2027-12-31	2027-12-31T19:00:00	2028-01-01T05:00:00+10:00	Io	-47.479	18.756

Saturn Moons Relative Offsets (sample)

system	date	datetime_utc	datetime_local	moon	dra_arcsec	ddec_arcsec
Saturn	2027-01-03	2027-01-03T00:00:00	2027-01-03T10:00:00+10:00	Enceladus	33.332	-0.9
Saturn	2027-01-03	2027-01-03T02:00:00	2027-01-03T12:00:00+10:00	Enceladus	27.284	0.756
Saturn	2027-01-03	2027-01-03T04:00:00	2027-01-03T14:00:00+10:00	Enceladus	17.35	2.34
Saturn	2027-01-03	2027-01-03T06:00:00	2027-01-03T16:00:00+10:00	Enceladus	4.895	3.564
Saturn	2027-01-03	2027-01-03T08:00:00	2027-01-03T18:00:00+10:00	Enceladus	-8.243	4.284
Saturn	2027-01-03	2027-01-03T10:00:00	2027-01-03T20:00:00+10:00	Enceladus	-20.193	4.356
Saturn	2027-01-03	2027-01-03T12:00:00	2027-01-03T22:00:00+10:00	Enceladus	-29.264	3.852
Saturn	2027-01-03	2027-01-03T14:00:00	2027-01-04T00:00:00+10:00	Enceladus	-34.159	2.772
Saturn	2027-01-03	2027-01-03T16:00:00	2027-01-04T02:00:00+10:00	Enceladus	-34.195	1.332
Saturn	2027-01-03	2027-01-03T18:00:00	2027-01-04T04:00:00+10:00	Enceladus	-29.3	-0.36
Saturn	2027-01-03	2027-01-03T20:00:00	2027-01-04T06:00:00+10:00	Enceladus	-20.193	-1.944
Saturn	2027-01-03	2027-01-03T22:00:00	2027-01-04T08:00:00+10:00	Enceladus	-8.135	-3.276
Saturn	2027-01-03	2027-01-04T00:00:00	2027-01-04T10:00:00+10:00	Enceladus	5.075	-4.104
Saturn	2027-01-03	2027-01-03T00:00:00	2027-01-03T10:00:00+10:00	Tethys	-40.711	4.284
Saturn	2027-01-03	2027-01-03T02:00:00	2027-01-03T12:00:00+10:00	Tethys	-42.942	3.276
Saturn	2027-01-03	2027-01-03T04:00:00	2027-01-03T14:00:00+10:00	Tethys	-41.862	2.052
Saturn	2027-01-03	2027-01-03T06:00:00	2027-01-03T16:00:00+10:00	Tethys	-37.651	0.684
Saturn	2027-01-03	2027-01-03T08:00:00	2027-01-03T18:00:00+10:00	Tethys	-30.524	-0.756
Saturn	2027-01-03	2027-01-03T10:00:00	2027-01-03T20:00:00+10:00	Tethys	-21.093	-2.16
Saturn	2027-01-03	2027-01-03T12:00:00	2027-01-03T22:00:00+10:00	Tethys	-10.007	-3.348

Minor Planet Opportunities (sample)

target	best_datetime_utc	best_datetime_local	best_altitude_deg	best_apmag	solar_elong_deg	score
3;	2027-12-31T12:00:00	2027-12-31T22:00:00+10:00	37.98	7.917	151.717	5
15;	2027-07-01T12:00:00	2027-07-01T22:00:00+10:00	86.03	9.74	153.136	5
7;	2027-06-15T12:00:00	2027-06-15T22:00:00+10:00	84.2	9.966	155.191	5
4;	2027-01-01T12:00:00	2027-01-01T22:00:00+10:00	29.96	7.879	95.637	4
1;	2027-12-27T00:00:00	2027-12-27T10:00:00+10:00	51.9	8.775	60.164	4
2;	2027-11-02T00:00:00	2027-11-02T10:00:00+10:00	49.9	8.87	60.335	3

Comet Opportunities (sample)

target	best_datetime_utc	best_datetime_local	best_altitude_deg	best_apmag	solar_elong_deg	score	rating
100P					0	0	unsc
101P					0	0	unsc
101P-B					0	0	unsc
102P					0	0	unsc
103P					0	0	unsc
104P					0	0	unsc

target	best_datetime_utc	best_datetime_local	best_altitude_deg	best_apmag	solar_elong_deg	score	rating
105P						0	unseen
106P						0	unseen
108P						0	unseen
109P						0	unseen
10P						0	unseen
110P						0	unseen
111P						0	unseen
112P						0	unseen
113P						0	unseen
114P						0	unseen
115P						0	unseen
116P						0	unseen
117P						0	unseen
118P						0	unseen

Lunar Occultations (sample)

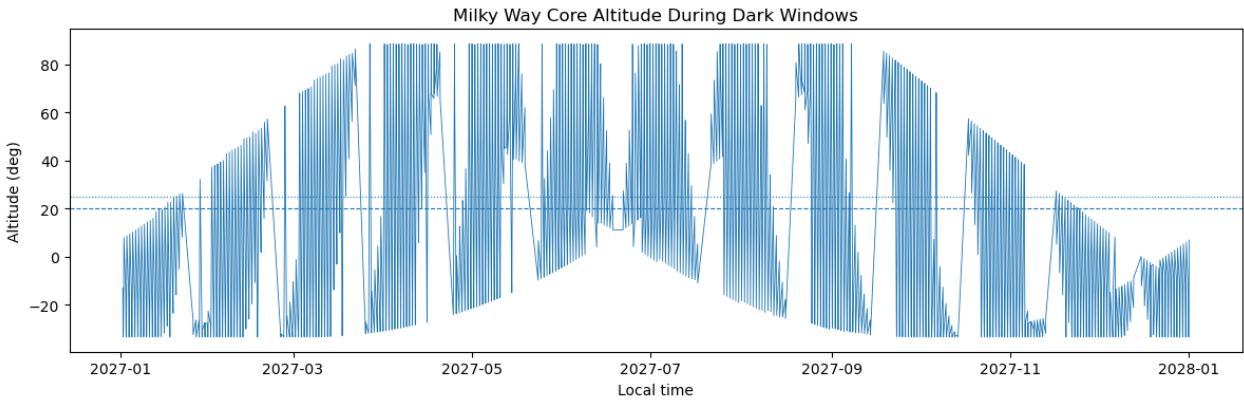
datetime_local	target	target_mag	moon_altitude_deg	limb	event_type	is_planetary	score
2027-05-01T20:15:00+10:00	Jupiter	-2.1	45.0	dark	planetary	True	excellent
2027-06-14T22:40:00+10:00	SAO 12345	7.8	32.0	bright	star	False	fair
2027-07-09T19:55:00+10:00	SAO 88888	10.7	18.0	dark	star	False	good

Meteor Showers (sample)

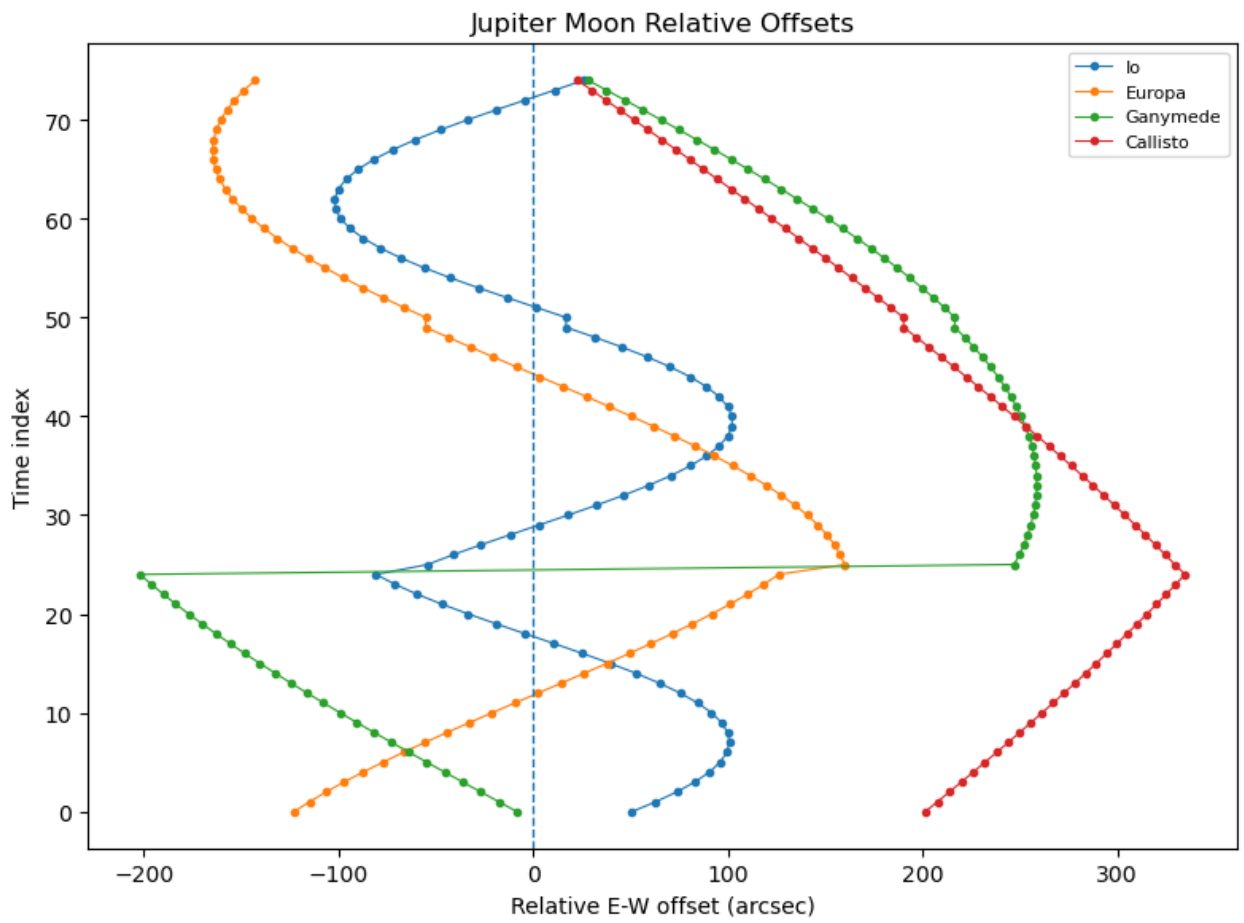
id	name	peak_date_local	radiant_alt_peak_deg	radiant_alt_predawn_deg	moon_illumination
quadrantids	Quadrantids	2027-01-03	-60.18	-6.01	0.272
eta_aquariids	Eta Aquariids	2027-05-06	-43.2	33.6	0.0
perseids	Perseids	2027-08-12	-35.52	1.97	0.763
orionids	Orionids	2027-10-21	-9.14	47.39	0.748

id	name	peak_date_local	radiant_alt_peak_deg	radiant_alt_predawn_deg	moon_illumination
geminids	Geminids	2027-12-14	9.71	22.71	1.0

milky_way_windows.png



strip_chart.png



strip_chart.png

